

Name: _____

Date: _____

FUNCTIONS AND GRAPHS, (TRANSFORMATIONS OF GRAPHS)

LO: I can describe and apply translations, reflections and stretches to graphs of functions.

Graph transformations

A transformation changes the position or shape of a graph.
Key types: translations (shifts), reflections (flips), and stretches (scaling).

Notation: $f(x) + a$ shifts up, $f(x + a)$ shifts left, $-f(x)$ reflects in x-axis, $f(-x)$ reflects in y-axis.

Transformation rules

●	$f(x) + 3$ shifts the graph up 3	vertical translation
●	$f(x - 2)$ shifts the graph right 2	horizontal translation
●	$-f(x)$ reflects in the x-axis	y-values change sign
●	$f(x + 2)$ shifts the graph left 2	it shifts RIGHT 2
●	$f(x) + 3$ shifts the graph up 3	it shifts UP 3

Key idea

Changes inside the bracket affect x (horizontal, opposite direction). Changes outside affect y (vertical, same direction).

$$y = af(x + b) + c$$

a = vertical stretch, b = horizontal shift (opposite), c = vertical shift

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - vertical translation

Describe the transformation from $y = x^2$ to $y = x^2 + 5$

Compare: $f(x) = x^2$ and $f(x) + 5$
The +5 is outside: vertical shift
Translation 5 units up

Translation by vector (0, 5)

Adding outside the function shifts the graph up.

Example 2 - horizontal translation

Describe the transformation from $y = x^2$ to $y = (x - 3)^2$

Compare: $f(x) = x^2$ and $f(x - 3)$
The -3 is inside: horizontal shift
Opposite direction: shift right 3

Translation by vector (3, 0)

Inside the bracket, the shift is in the opposite direction.

Example 3 - reflection in x-axis

Describe the transformation from $y = x^3$ to $y = -x^3$

Compare: $f(x) = x^3$ and $-f(x)$
The negative is outside: reflects in x-axis
Every y-value changes sign

Reflection in the x-axis

All positive y-values become negative and vice versa.

Example 4 - vertical stretch

Describe the transformation from $y = \sin x$ to $y = 3 \sin x$

Compare: $f(x) = \sin x$ and $3f(x)$
The 3 multiplies outside: vertical stretch
Scale factor 3 in the y-direction

Vertical stretch, scale factor 3

Every y-coordinate is multiplied by 3.

YOUR TURN – PRACTICE (deliberate practice)

1. Vertical translations

Describe each transformation from $y = f(x)$.

- a) $y = f(x) + 4$
- b) $y = f(x) - 2$
- c) $y = x^2 + 7$ from $y = x^2$
- d) $y = x^3 - 1$ from $y = x^3$

2. Horizontal translations

Describe each transformation from $y = f(x)$.

- a) $y = f(x - 5)$
- b) $y = f(x + 3)$
- c) $y = (x - 1)^2$ from $y = x^2$
- d) $y = (x + 4)^3$ from $y = x^3$

3. Reflections

Describe each transformation.

- a) $y = -f(x)$
- b) $y = f(-x)$
- c) $y = -x^2$ from $y = x^2$
- d) $y = (-x)^3$ from $y = x^3$

4. Stretches

Describe the stretch.

- a) $y = 2f(x)$
- b) $y = f(3x)$
- c) $y = 5 \sin x$ from $y = \sin x$
- d) $y = \cos(2x)$ from $y = \cos x$

5. Mixed transformations

Describe the transformation from $y = f(x)$.

- a) $y = f(x + 2) + 3$
- b) $y = -f(x) + 1$
- c) $y = 2f(x - 4)$
- d) $y = f(2x) - 5$
- e) $y = -f(x - 1) + 2$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $f(x + 2)$ shifts the graph left 2

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Correct answer: _____

Reason: _____

b) $f(x + 2)$ shifts the graph right 2

☐

Correct answer: _____

Reason: _____

c) $-f(x)$ reflects the graph in the x-axis

☐

Correct answer: _____

Reason: _____

d) $f(x) + 3$ shifts the graph right 3

☐

Correct answer: _____

Reason: _____

e) $2f(x)$ stretches the graph vertically by factor 2

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Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The point **(2, 5)** lies on $y = f(x)$. Find the coordinates of the corresponding point on: (i) $y = f(x) + 3$, (ii) $y = f(x - 4)$, (iii) $y = 2f(x)$, (iv) $y = -f(x)$.

Expression: _____

Simplified: _____

b) The vertex of a parabola $y = x^2$ is at the origin. After two transformations the vertex is at **(3, -2)**. Write two possible equations and describe the transformations applied.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

- Have I shown all my working clearly?
- Did I check my answer using a different method?
- Does my answer look reasonable?

